



SAKET KUMAR

Data Scientist • AI/ML Engineer • LLM Systems

BSc (Hons.) Data Science & AI • Indian Institute of Technology,
Guwahati

+91-9693899769

Email

Portfolio

[GitHub](#) • [LinkedIn](#) • [Portfolio](#) • [Gmail](#)

Summary

IIT Guwahati Data Science & AI undergraduate (CGPA 8.4/10) who **ships**: 15+ end-to-end AI systems — trained, benchmarked, tested, and deployed live — spanning LLM agents & RAG, computer vision, speech AI, and on-device WASM inference. Strength is end-to-end delivery: model training, quantitative evaluation (F1, AUC, WER, ANLS), CI-tested code, and production deployment. Every claim below links to a public repository or live application.

Education

BSc (Hons.) in Data Science and Artificial Intelligence **Indian Institute of Technology, Guwahati** **2023 – Present**

CGPA: 8.4/10 | Coursework: Machine Learning, Data Mining & Warehousing, Statistical Inference, Time Series Forecasting, Recommender Systems, Optimization, Econometrics, Algorithm Design & Analysis, RDBMS

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, R, C, Bash

ML & Deep Learning: PyTorch, Scikit-learn, XGBoost, TensorFlow/Keras, OpenCV, Hugging Face Transformers

Generative AI / LLM Systems: RAG (BM25 retrieval + NLI verification), LLM agents & multi-agent orchestration, Whisper, sherpa-onnx, LLM evaluation & safety gates

On-Device / Edge AI: ONNX Runtime, WebAssembly, int8 quantization, fully in-browser inference

Full-Stack & Cloud: React, Next.js, FastAPI, Node.js, Three.js, GitHub Actions (CI), Docker, Vercel / Netlify / Render

Data & Tools: Pandas, NumPy, Spark, Hadoop, Git, Linux, Remote Sensing (Sentinel-1 SAR), GIS

Projects

RepoLens — LLM Code-Review Agent

[GitHub](#)

Python • LLM Agent • BM25 + Reference-Graph Retrieval • GitHub Actions Autonomous code reviewer that reads a PR diff *plus the entire repository* via BM25 and reference-graph retrieval, then posts line-level review comments on GitHub pull requests. Shipped with **65 passing tests**, green CI, and a CI-friendly exit-code protocol (approve / comment / request-changes).

FaithGuard — Hallucination Detection & Mitigation for RAG

[GitHub](#)

Python • NLI • HistGBM • RAG • FastAPI • Docker Claim-level hallucination detector for RAG answers: a learned 15-feature classifier (HistGBM) over NLI features trained on 900 labeled examples, a hard contradiction safety floor, and auto-mitigation via claim-guided re-retrieval and corrective regeneration. Detection **F1 0.887 / AUC 0.948** (+13.7 F1 over the NLI-threshold baseline); end-to-end faithfulness 0.497 → 0.620 on a 120-question held-out evaluation.

Deepfake Detection — Spatial-Temporal Video Forensics

[GitHub](#)

PyTorch • EfficientNet-B0 • Transformer • FF++ / Celeb-DF v2 Spatial (EfficientNet-B0) + temporal transformer detector for manipulated video, evaluated with a length-controlled protocol on leakage-free splits, trained CPU-only. Test video **AUC 0.9553** on FaceForensics++; **0.9220 cross-source** on Celeb-DF v2 (out-of-distribution generators) — 0.9033 overall across 5,486 held-out videos.

Meeting Intelligence — In-Browser Speech-to-Actions Pipeline

[GitHub](#) • [Live Demo](#)

Whisper • *sherpa-onnx (WASM)* • *Diarization* • *FastAPI* • *Vercel* Fully in-browser pipeline: int8 pyannote segmentation → ERes2Net speaker embeddings → clustering → Whisper ASR → LLM analysis, producing summaries, decisions, and owner-assigned action items with zero audio leaving the device. **WER 7.81%**, action-item **F1 0.947** with 100% speaker attribution on a labeled 3-speaker benchmark; deployed live.

Medical Report Summarizer — Clinical NLP with Safety Gate

[GitHub](#) • [Live Demo](#)

Clinical NLP • *Safety-Gated LLM* • *FastAPI* • *Vercel* Patient-facing radiology & pathology report summarizer: a deterministic triage core decodes clinical jargon; a safety-gated LLM layer rewrites only after passing a faithfulness gate. **100% triage accuracy (47/47 cases)**, 100% must-flag recall (41/41 critical findings), zero false alarms on negation-safety evaluation; deployed live.

DocVQA — Visual Question Answering over Documents

[GitHub](#) • [Live Demo](#)

Python • *OCR* • *Layout Analysis* • *BM25 Retrieval* • *FastAPI* Visual question answering over documents: OCR + layout analysis + BM25 retrieval, answered by a grounded LLM. **ANLS 0.981** over 27 ground-truth questions driven through the live deployed app in a headless browser; every question the models answered scored a perfect 1.000.

Svara — On-Device Speech Emotion Recognition

[GitHub](#) • [Live Demo](#)

PyTorch • *ONNX* • *WebAssembly* • *CREMA-D* Privacy-first 6-class speech-emotion screener: a 287K-parameter CNN quantized to int8 ONNX (~0.3 MB), running fully in-browser via WebAssembly — no audio ever leaves the device. Speaker-disjoint test accuracy 0.596 ($\approx 3.5\times$ chance level), inference under 200 ms on laptop WASM.

Yamuna Flood Mapper — Sentinel-1 SAR Flood Mapping

[GitHub](#) • [Live Demo](#)

Sentinel-1 • *SAR Remote Sensing* • *GIS* • *Vercel* Flood-extent mapping for the Yamuna corridor, Delhi built from free Sentinel-1 SAR satellite imagery, delivered as a live early-warning flood layer for the city.

Vridhhi — Self-Improving Paper-Trading Agent

[GitHub](#) • [Live Demo](#)

Python • *Multiplicative-Weights Hedge Allocator* • *Walk-Forward ML* • *Risk Overlay* Adaptive trading agent: a windowed Hedge meta-allocator (with regret bound) over 7 decorrelated strategies, a walk-forward ML predictor with validation gate, and an adaptive risk overlay that meta-learns its own learning rate. In a 5.5-year walk-forward backtest with fees, the risk overlay cuts max drawdown from **−91% (buy & hold) to −21%** at comparable Sharpe (0.88), including a no-lookahead out-of-time holdout.

Achievements

- Built and **deployed 15+ open-source AI systems** end-to-end — every flagship shipped with quantitative benchmarks (F1, AUC, WER, ANLS), automated tests, CI, and a live public deployment.
- **15+ public repositories** on GitHub with production-grade code, tests and documentation; interactive 3D portfolio showcasing the full build journey.
- Cross-source deepfake detection research — detector generalizes **0.92 AUC** to unseen generator families (Celeb-DF v2), addressing the field's hardest open problem.
- **CGPA 8.4/10** at IIT Guwahati — 6 trimesters of rigorous DS & AI coursework completed.
- **More builds:** Hinglish ASR (code-switched Hindi–English) • Persona (E2E-encrypted ephemeral chat) • MindVault (local-first AI knowledge assistant) • Lost & Found Network • ADA God's Eye (3D live globe: flights, satellites, world events) — all on [GitHub](#).